DAEN 427 Homework 1

Due date: Aug 26, 2025

1. Explain the difference between a *confounder* and a *collider* in the context of a DAG. Why is it important not to condition on a collider when trying to estimate a causal effect?

   Answer:

   (a) Confounders are common causes of both the exposure (A) and outcome (Y).

   (b) Colliders are common effects (i.e. they are caused by two variables).

   (c) Conditioning on confounders helps reduce bias, while conditioning on colliders can introduce bias by opening a backdoor path.

2. You are studying whether a new training program (A) improves job performance (Y). Based on domain knowledge, age and prior experience influence both program participation and job performance. Gender is believed to influence only program participation. Construct a DAG using this information, and identify the minimum set of covariates you would need to adjust for to estimate the causal effect of A on Y.

   Answer:

   (a) Consider a directed acyclic graph (DAG) with the following structure:

   - $A \to Y$
   - Age $\to A$ and Age $\to Y$
   - Experience $\to A$ and Experience $\to Y$
   - Gender $\to A$

   This DAG represents the assumed causal relationships among the variables: Age, Experience, and Gender influence the treatment variable A, and both Age and Experience also directly affect the outcome Y. The treatment A has a direct causal effect on the outcome Y.

   (b) Adjustment set: Age and Experience (not Gender).

3. Refer to the "Uncomplicated Appendicitis" DAG example in the slides. Discuss the role of *income* and *primary language* in this model. Which one is unmeasured? What potential biases could arise from not including it in your analysis?

   Answer:

   (a) Income is marked as an unmeasured variable (U), primary language as measured.

   (b) Not including income could leave an open backdoor path between treatment and outcome, leading to confounding bias.

   (c) DAGs help visualize and assess such risks before analysis.

4. Consider a DAG where a treatment variable A causally affects an outcome Y, and there exists a confounder C that influences both A and Y.

   (a) Explain why the conditional probability $P(Y \mid A)$ is not equal to the causal effect $P(Y \mid \text{do}(A))$.

   (b) Describe how adjusting for C allows us to estimate the causal effect using observational data.

   Answer:

   - The expression $P(Y \mid A)$ includes confounding bias because it captures associations from both causal and non-causal paths (e.g., through C).
   - The causal effect $P(Y \mid \text{do}(A))$ represents the distribution of Y if we were to intervene and set A, breaking the natural causes of A.

     Remember, in probability theory $P(Y \mid A)$ means "the probability of Y given that we observed A." In causal inference, $P(Y \mid \text{do}(A))$ means "the probability of Y if we intervene and set A to a specific value." This is a causal statement, not just a statistical one.
   - Adjustment for C blocks the backdoor path from A to Y, isolating the causal effect.

- Using the backdoor adjustment formula:

$$P(Y \mid \mathsf{do}(A)) = \sum_c P(Y \mid A, C = c) \cdot P(C = c)$$